# Exercise 4

## 1. SQL JOINS

### 1. INNER JOIN

```
SELECT
        E.EmployeeID,
        E.FirstName,
        E.LastName,
        E.Department,
        E.Salary,
        P.ProjectID,
        P.ProjectName,
        P.Budget,
        P.Status
FROM    Employees AS E
INNER JOIN  Projects AS P
ON      E.EmployeeID = P.EmployeeID;
```

| EmployeeID | FirstName | LastName | Department | Salary | Project ID | ProjectName | Budget | Status |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 | 101 | AI Development | 100000 | completed |
| ~~1~~ | John | Doe | IT | 70000 | 103 | Cybersecurity Audit | 75000 | Pending |
| ~~3~~ 2 | Alice | Smith | HR | 60000 | 102 | Employee Training | 50000 | ongoing |
| 3 | Bob | Johnson | Finance | 75000 | 104 | Financial Analysis | 90000 | ongoing |
| 5 | Emma | Wilson | Sales | 65000 | 105 | Market Expansion | 65000 | completed |
| 6 | Michael | Clark | Finance | 80000 | 106 | Risk management | 80000 | Pending |

2. LEFT JOIN

```
SELECT
        E.EmployeeID,
        E.FirstName,
        E.LastName,
        E.Department,
        E.Salary,
        P.ProjectID,
        P.ProjectName,
        P.Budget,
        P.Status
FROM    Employees AS E
LEFT JOIN  Projects AS P
ON      E.EmployeeID = P.EmployeeID;
```

| EmployeeID | FirstName | LastName | Department | Salary | ProjectID | ProjectName | Budget | Status |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 | 101 | AI Development | 100000 | completed |
| 1 | John | Doe | IT | 70000 | 103 | Cybersecurity Audit | 75000 | Pending |
| 2 | Alice | Smith | HR | 60000 | 102 | Employee Training | 50000 | ongoing |
| 3 | Bob | Johnson | Finance | 75000 | 104 | Financial Analysis | 90000 | ongoing |
| 4 | David | Brown | IT | 72000 | NULL | NULL | NULL | NULL |
| 5 | Emma | Wilson | Sales | 65000 | 105 | Market Expansion | 65000 | completed |
| 6 | Michael | Clark | Finance | 80000 | 106 | Risk Management | 50000 | Pending |

3. RIGHT JOIN

```
SELECT
        P.ProjectID,
        P.ProjectName,
        P.Budget,
        P.Status,
        E.EmployeeID,
```

```
    E.FirstName,
    E.LastName,
    E.Department,
    E.Salary
FROM    Employees AS E
RIGHT JOIN    Projects AS P
    ON    E.EmployeeID = P.EmployeeID;
```

| ProjectID | ProjectName | Budget | Status | EmployeeID | FirstName | LastName | Department | Salary |
|---|---|---|---|---|---|---|---|---|
| 101 | AI Development | 100000 | completed | 1 | John | Doe | IT | 70000 |
| 102 | Employee Training | 50000 | Ongoing | 2 | Alice | Smith | HR | 60000 |
| 103 | cybersecurity Audit | 75000 | Pending | 1 | ~~Bob~~ John | Doe | IT | 70000 |
| 104 | Financial Analysis | 90000 | Ongoing | 3 | Bob | Johnson | Finance | 75000 |
| 105 | Market Expansion | 65000 | Completed | 5 | Emma | Wilson | Sales | 65000 |
| 106 | Risk Management | 80000 | Pending | 6 | Michael | Clark | Finance | 80000 |

4 <u>FULL OUTER JOIN</u>

```
SELECT
    E.EmployeeID,
    E.FirstName,
    E.LastName,
    E.Department,
    E.Salary,
    P.ProjectID,
    P.ProjectName,
    P.Budget,
    P.Status
FROM    Employees AS E
FULL OUTER JOIN    Projects AS P
ON    E.EmployeeID = P.EmployeeID
```

| EmployeeID | FirstName | LastName | Department | Salary | ProjectID | ProjectName | Budget | Status |
|---|---|---|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 | 101 | AI Development | 100000 | Completed |
| 1 | John | Doe | IT | 70000 | 103 | Cybersecurity Audit ~~Employee Training~~ | 75000 | Pending |
| 2 | Alice | Smith | HR | 60000 | 102 | Employee Training | 50000 | Ongoing |
| 3 | Bob | Johnson | Finance | 75000 | 104 | Financial Analysis | 90000 | Ongoing |
| 4 | David | Brown | IT | 72000 | Null | Null | Null | Null |
| 5 | Emma | Wilson | Sales | 65000 | 107 | Market Expansion | 65000 | Completed |
| 6 | Michael | Clark | Finance | 80000 | 106 | Risk Management | 80000 | Pending |

## 3. Filtering Statements

7.
```
SELECT    EmployeeID,
          FirstName,
          LastName,
          Department,
          Salary
FROM      employees
WHERE     salary > 70000 ;
```

| EmployeeID | FirstName | LastName | Department | Salary |
|---|---|---|---|---|
| 3 | Bob | Johnson | Finance | 75000 |
| 4 | David | Brown | IT | 72000 |
| 6 | Michael | Clark | Finance | 80000 |

8.
```
SELECT    EmployeeID,
          FirstName,
          LastName,
          Department,
          Salary
FROM      employee
WHERE     departments IN('IT', 'Finance');
```

| EmployeeID | FirstName | LastName | Department | Salary |
|---|---|---|---|---|
| 1 | John | Doe | IT | 70000 |
| 3 | Bob | Johnson | Finance | 75000 |
| 4 | David | Brown | IT | 72000 |
| 6 | Michael | Clark | Finance | 80000 |

9. SELECT ProjectID,
ProjectName,
Budget,
Status
FROM Projects
WHERE Status != 'completed';

| ProjectID | ProjectName | Budget | Status |
|---|---|---|---|
| 103 | Employee Training | 50000 | Ongoing |
| 103 | Cybersecurity Audit | 75000 | Pending |
| 104 | Financial Analysis | 90000 | Ongoing |
| 106 | Risk Management | 80000 | Pending |

10. SELECT ProjectID,
ProjectName,
Budget,
Status
FROM Projects
WHERE budget > 70 000
AND status != 'completed';

| ProjectID | ProjectName | Budget | Status |
|---|---|---|---|
| 103 | Cybersecurity Audit | 75000 | Pending |
| 104 | Financial Analysis | 90000 | Ongoing |
| 106 | Risk Management | 80000 | Pending |

11.
```
SELECT      EmployeeID,
            FirstName,
            LastName,
            Department,
            Salary,
            City
FROM        Employees
WHERE       City IN ('New York', 'Toronto')
ORDER BY    salary DESC;
```

| EmployeeID | FirstName | LastName | Department | salary | City |
|---|---|---|---|---|---|
| 6 | Michael | Clark | Finance | 80000 | New York |
| 3 | Bob | Johnson | Finance | 75000 | Toronto |
| 1 | John | Doe | IT | 70000 | New York |

12.
```
SELECT      EmployeeID,
            FirstName,
            LastName,
            Department,
            Salary
FROM        Employees
ORDER BY    salary DESC
LIMIT       3;
```

| EmployeeID | First-Name | LastName | Department | salary |
|---|---|---|---|---|
| 6 | Michael | Clark | Finance | 80000 |
| 3 | Bob | Johnson | Finance | 75000 |
| 4 | David | Brown | IT | 72000 |

## 4. Aggregate Functions with Group BY & HAVING

13.
```
SELECT    Department,
          SUM(salary) AS TotalSalary
FROM      Employees
GROUP BY  Department
ORDER BY  TotalSalary DESC;
```

| Department | TotalSalary |
|---|---|
| Finance | 155000 |
| IT | 142000 |
| Sales | 65000 |
| HR | 60000 |

14.
```
SELECT    City,
          AVG(salary) AS Averagesalary
FROM      Employees
GROUP BY  City
HAVING    AVG(salary) > 65000;
```

| City | Averagesalary |
|---|---|
| New York | 75000 |
| Toronto | 75000 |
| London | 72000 |

15.
```
SELECT    Department,
          COUNT(EmployeeID) AS EmployeeCount
FROM      Employees
GROUP BY  Department
HAVING    COUNT(EmployeeID) > 1;
```

| Department | Employee Count |
|---|---|
| IT | 2 |
| Finance | 2 |

~~16. SELECT Status, Project Count~~

16. SELECT Status,
COUNT (ProjectID) AS Project Count
FROM Projects
GROUP BY Status
HAVING COUNT (ProjectID) >= 2;

| Status | Project Count |
|---|---|
| Completed | 2 |
| Ongoing | 2 |
| Pending | 2 |

17. SELECT E.EmployeeID,
E.FirstName,
E.LastName,
SUM (P.Budget) AS TotalProjectBudget
FROM Employees AS E
JOIN Projects AS P
ON E.EmployeeID = P.EmployeeID
GROUP BY E.EmployeeID
E.FirstName
E.LastName
HAVING SUM (P.Budget) > 150000;

| EmployeeID | FirstName | LastName | TotalProjectBudget |
| --- | --- | --- | --- |
| 1 | John | Doe | 175000 |